

Isolation-Enhanced Compact Four-Element MIMO Antenna for 5G NR Millimeter-Wave and D2D Applications

Md. Najmul Hossain

Department of Electrical, Electronic and Communication Engineering

Pabna University of Science and Technology

Pabna-6600, Bangladesh

najmul_ece@pust.ac.bd

ORCID: 0000-0002-2528-9168

Sayed Shifat Ahmed

Department of Electrical and Electronic Engineering

RTM Al-Kabir Technical University

Sylhet-3100, Bangladesh

shikor99@gmail.com

ORCID: 0009-0007-9050-6951

Md. Sarowar Hossain

Department of Electrical, Electronic and Communication Engineering

Pabna University of Science and Technology

Pabna-6600, Bangladesh

mdsarowar1090@gmail.com

Tetsuya Shimamura

Graduate School of Science and Engineering

Saitama University

Saitama 338-8570, Japan

shima@mail.saitama-u.ac.jp

ORCID: 0000-0002-7874-3467

Abstract— In this paper, we present a compact four-element multiple-input multiple-output (MIMO) antenna with enhanced isolation for 5G New Radio (NR) millimeter-wave (mmWave) and device-to-device (D2D) communication applications. The antenna is implemented on a low-loss Rogers RT5880 substrate with a relative permittivity of 2.2 and a loss tangent of 0.0009. An optimized element arrangement is employed to suppress mutual coupling and improve overall electromagnetic performance. Simulation results show that the antenna operates over a wide frequency range from 32.198 to 53.327 GHz, providing an impedance bandwidth of 21.13 GHz below the -10 dB reflection coefficient. High isolation of up to -47.43 dB is achieved across the operating band, while stable radiation characteristics are maintained. The proposed antenna occupies a compact footprint of 36×36 mm², delivers a peak gain of 8.71 dB, and exhibits a radiation efficiency exceeding 92%. Furthermore, the envelope correlation coefficient remains on the order of 10^{-6} , with a minimum value of 1.2875×10^{-6} , indicating excellent diversity performance. Owing to these characteristics, the proposed design is well-suited for compact MIMO implementations in 5G NR mmWave and short-range D2D communication systems.

Keywords— MIMO antenna, millimeter-wave, 5G NR, D2D communication, isolation enhancement

I. INTRODUCTION

The rapid evolution of wireless communication technologies has driven the deployment of fifth-generation (5G) New Radio (NR) networks, where millimeter-wave (mmWave) frequency bands play a key role in enabling ultra-high data rates, low latency, and increased network capacity. To realize high-performance wireless links at these frequencies, compact antenna systems are required that occupy a limited physical area

while maintaining stable radiation characteristics and sufficient gain [1], [2].

Multiple-input multiple-output (MIMO) antenna technology is a cornerstone of modern high-capacity wireless systems, leveraging spatial diversity and multiplexing to enhance channel capacity, link reliability, and spectral efficiency [3]. However, integrating multiple radiating elements within a small aperture introduces mutual coupling, reduced isolation, distorted radiation patterns, and increased envelope correlation coefficient (ECC), collectively degrading MIMO performance [4]. These challenges are particularly pronounced at mmWave frequencies because of reduced electrical spacing and stronger electromagnetic interactions among antenna elements.

To mitigate coupling and correlation effects, several isolation-enhancement techniques have been investigated, including defected ground structures (DGS), electromagnetic bandgap (EBG) structures, and metamaterial-inspired decoupling networks [5]. Although several isolation-enhancement techniques have been reported for mmWave MIMO antennas, achieving a simultaneous trade-off among compact size, ultra-wide bandwidth, high isolation, low ECC, and high radiation efficiency remains a major challenge. In practical 5G NR handheld and short-range D2D communication systems, antenna elements are often placed within limited spacing constraints, which increases electromagnetic coupling and deteriorates diversity performance. Furthermore, many recently reported designs either require relatively large footprints, exhibit limited bandwidth, or provide insufficient isolation performance for dense mmWave integration scenarios. Therefore, there is a strong need for compact multi-element mmWave MIMO antennas capable of maintaining high isolation and low correlation while preserving wideband performance and

stable radiation characteristics. Motivated by these challenges, this work proposes a compact four-element MIMO antenna with integrated decoupling structures specifically optimized for 5G NR mmWave and D2D applications.

Several recent mmWave MIMO antenna designs have been reported for 5G NR applications with different trade-offs among bandwidth, gain, isolation, and compactness. Abbasi *et al.* [13] proposed an eight-element mm-wave MIMO antenna using modified circular patch radiators for 5G applications, achieving 2.6 GHz bandwidth, isolation above 25 dB, and a peak gain of 7.5 dBi; however, the design suffered from relatively narrow bandwidth and moderate gain performance. In [14] Zakeri *et al.* introduced a flower-shaped four-port MIMO antenna with an evolved defected ground structure that achieved 13.5 GHz bandwidth, 9.8 dBi gain, and isolation greater than 26 dB, although the incorporation of multiple CSRR and slot structures increased design complexity and fabrication sensitivity. Similarly, Khan *et al.* developed a compact T-shaped four-element MIMO antenna for 5G NR systems with 9.8 dBi gain, wideband performance, and efficiency up to 89%, but the achieved isolation of 18 dB remained comparatively lower than several recently reported high-isolation mm-wave MIMO antennas [15].

A number of recent mmWave MIMO antennas have been reported with different trade-offs among bandwidth, gain, isolation, ECC, and efficiency. Dual-band 28/38 GHz smartphone-oriented MIMO structures have been presented in the literature [6], and multi-band mmWave MIMO platforms covering 5G and legacy bands have also been reported [7]. Dual-band 28/38 GHz MIMO configurations with improved radiation characteristics have been investigated in [8]. Circularly polarized 28/38 GHz MIMO antennas have been explored to support polarization diversity [9]. More recently, pattern-diversity-based dual-band MIMO patch antennas have been proposed to improve spatial diversity performance in 5G mmWave scenarios [10]. Despite these advances, achieving broadband operation together with extremely low ECC, high isolation, and compact size remains challenging for mmWave MIMO antennas intended for 5G NR and D2D applications. This work addresses these challenges by proposing a compact four-element MIMO antenna operating over 32.198–53.327 GHz is proposed for 5G NR mmWave and D2D applications including n260 and n262 [11]. A defected ground structure combined with orthogonal decoupling strips is introduced to significantly suppress mutual coupling among antenna elements. The proposed design achieves high isolation of up to –47.43 dB and an ultra-low ECC of 1.2875×10^{-6} while maintaining approximately 92% radiation efficiency. The antenna maintains stable radiation characteristics and a compact footprint of only $36 \times 36 \text{ mm}^2$, making it suitable for compact mmWave wireless devices.

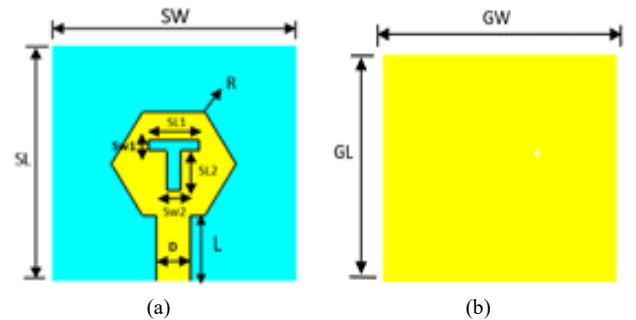
II. ANTENNA DESIGN AND CONFIGURATION

The primary design objective is to achieve a compact multi-element antenna structure capable of wideband mmWave operation while minimizing mutual coupling among closely spaced antenna elements. To address this problem, the proposed design employs a combination of geometrical slot modifications and defected ground plane techniques. Initially, the radiating

patch geometry is optimized to improve impedance bandwidth and radiation characteristics. Subsequently, ground-plane decoupling structures, including a longitudinal slot and orthogonal isolation strips, are introduced to suppress surface-wave propagation and reduce electromagnetic interaction between adjacent elements. Parametric optimization is then performed to achieve an appropriate balance among impedance matching, isolation, gain, ECC, and radiation efficiency. The simulations were carried out using CST Microwave Studio based on the finite integration technique. The antenna structure was modeled using copper conductors with a conductivity of $5.8 \times 10^7 \text{ S/m}$, while low dielectric loss Rogers RT5880 was used as the substrate material with a relative permittivity of 2.2 and a loss tangent of 0.0009. Open boundary conditions and adaptive meshing were employed to ensure accurate far-field and scattering-parameter analysis across the operating frequency range. The antenna is excited through a microstrip feed line of width 2.454 mm and length 5.12 mm, which is designed to ensure proper impedance matching and efficient power transfer. The radiating element has a circular geometry with an outer radius of 4.5 mm and is centrally positioned at coordinates (9, 9) mm on the substrate. The front and back views of the single-element antenna are illustrated in Fig. 1(a) and Fig. 1(b), respectively. In full-wave electromagnetic simulations, the curved patch geometry is approximated using a segmented structure composed of six sections to accurately model the circular shape.

The proposed antenna is implemented on a Rogers RT5880 substrate, selected for its low dielectric loss and stable electrical characteristics at millimeter-wave frequencies. The single-element antenna is fabricated on a substrate with dimensions of $18 \times 18 \times 0.79 \text{ mm}^3$, a relative permittivity of 2.2, and a loss tangent of 0.0009. The radiating patch and the microstrip feed line are etched on the top surface of the substrate, while a continuous copper ground plane is placed on the bottom side. Both the radiating patch and the ground plane are realized using copper with a thickness of 0.035 mm.

The antenna is excited via a microstrip feed line with a width of 2.454 mm and a length of 5.12 mm, designed to ensure proper impedance matching and efficient power transfer. The radiating element has a circular geometry with an outer radius of 4.5 mm and is centrally positioned at coordinates (9, 9) mm on the substrate. The front and back views of the single-element antenna are illustrated in Fig. 1(a) and Fig. 1(b), respectively. In full-wave electromagnetic simulations, the curved patch geometry is approximated using a segmented structure composed of six sections to accurately model the circular shape.



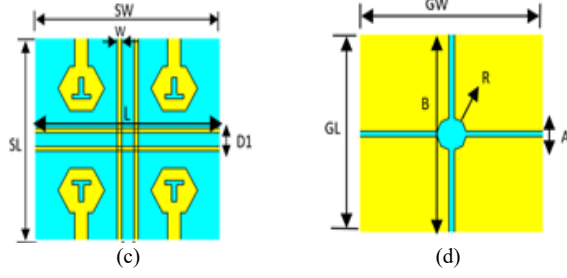


Fig. 1. (a) Front view of single-element, (b) back view of single-element, (c) Front view of 4-element, and (d) Front view of 4-element.

To enhance impedance bandwidth and improve surface current distribution, two rectangular slots are introduced into the radiating patch. Slot-1 has dimensions of $3.6 \times 0.8 \text{ mm}^2$, while Slot-2 measures $3 \times 1 \text{ mm}^2$. The detailed geometrical parameters of the single-element antenna are summarized in Table I.

TABLE I. OPTIMIZED GEOMETRICAL PARAMETERS OF THE SINGLE-ELEMENT ANTENNA

Parameter	Value (mm)	Parameter	Value (mm)
$S_L = S_W$, (Substrate length=width)	18	SW_2 , (Slot-2 width)	1.0
R , (Patch outer radius)	4.5	L , (Feed line length)	5.12
SL_1 , (Slot-1 length)	3.6	D , (Feed line width)	2.454
SW_1 , (Slot-1 width)	0.8	G_L , (Ground length)	18
SL_2 , (Slot-2 length)	3.0	G_W , (Ground width)	18

The four-element MIMO configuration is formed by arranging identical patch elements in a 2×2 array on a Rogers RT5880 substrate with overall dimensions of $36 \times 36 \text{ mm}^2$. The front and rear views of the four-element antenna structure are shown in Fig. 1(c) and Fig. 1(d), respectively. Each antenna element is independently excited through its own feed line and matching network to ensure proper operation and minimize interaction between ports.

To improve isolation among the antenna elements, several decoupling features are incorporated into the ground plane. A longitudinal slot of dimensions $36 \times 1 \text{ mm}^2$ is etched into the ground plane to suppress surface current coupling. In addition, two orthogonally intersecting decoupling strips with a width of 0.8 mm and a spacing of 2.4 mm are introduced at the center of the ground plane, effectively limiting current propagation

between adjacent elements. Furthermore, segmented circular slots with a radius of 3 mm are etched into the ground plane to provide additional isolation enhancement. The dimensions of the complete 4-port MIMO antenna structure are optimized through parametric analysis and are listed in Table II. These four antenna elements are designated as Port-1, Port-2, Port-3, and Port-4 in clockwise order, starting from the upper-left element.

These design features collectively enable high isolation while maintaining a compact footprint, high radiation efficiency, and stable performance across the targeted 5G NR millimeter-wave and D2D communication bands.

TABLE II. OPTIMIZED GEOMETRICAL PARAMETERS OF THE FOUR-ELEMENT MIMO ANTENNA

Parameter	Value (mm)	Parameter	Value (mm)
S_W , (Substrate width)	36	G_L , (Ground length)	36
S_L , (Substrate length)	36	G_W , (Ground width)	36
L , (Isolation length)	36	B , (Ground slot length)	36
W , (Isolation width)	0.8	A , (Ground slot width)	1
D_1 , (Distance between parallel isolations)	2.4	R , (Ground 8-segment cylindrical slot radius)	3

III. RESULTS AND ANALYSIS

The performance of the proposed antenna is evaluated through full-wave electromagnetic simulations, focusing on key parameters including reflection coefficient, envelope correlation coefficient (ECC), gain, directivity, radiation efficiency, input impedance, and far-field characteristics. The simulated reflection coefficient (S_{11}) is shown in Fig. 2 (a). The antenna resonates at 33.33 GHz, where a minimum return loss of -35.33 dB is observed. The reflection coefficient remains below -10 dB , resulting in an impedance bandwidth of 21.13 GHz. On the other side, mutual coupling between antenna elements in Fig. 2 (b) shows $< -20 \text{ dB}$, considered good isolation between multiple antenna elements in the context of MIMO configuration. Across the frequency range from 32.198 GHz to 53.327 GHz. This low level of coupling ensures each antenna element operates independently, and its wide operating bandwidth covers the major 5G NR millimeter-wave bands. Fig. 2 (c) presents the simulated gain and directivity as functions of frequency. At the resonant frequency of 33.33 GHz, the antenna achieves a peak gain of 8.71 dB and a peak directivity of 9.24 dBi.

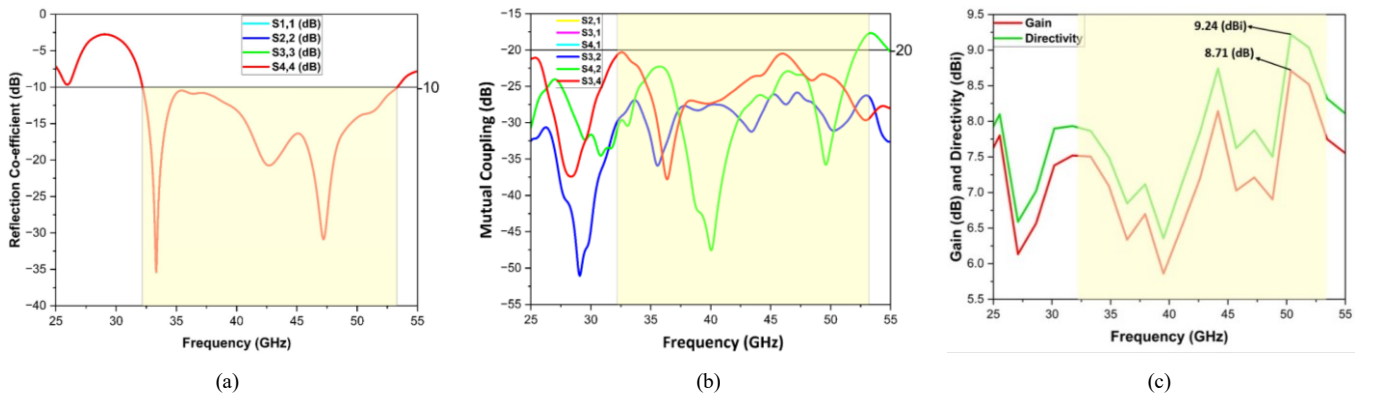


Fig. 2. Simulated graph of (a) reflection coefficient, (b) mutual coupling, and (c) gain and directivity analysis of a MIMO antenna.

These values correspond to a radiation efficiency of approximately 92%, as shown in Fig. 3, indicating that the majority of the input power is effectively radiated rather than dissipated as loss. Both gain and directivity remain relatively stable over the primary operating band, ensuring consistent far-field performance. The three-dimensional far-field radiation patterns at the resonant frequency are illustrated in Fig. 4 (a) and Fig. 4 (b), providing a clear visualization of the antenna's directional characteristics and beam shape.

The relationship between gain and directivity is given by

$$G = \eta_{tot} D \quad (1)$$

where η_{tot} denotes the total radiation efficiency and D represents the antenna directivity.

Further insight into the radiation behavior is obtained from the far-field E-plane and H-plane patterns shown in Fig. 5, corresponding to the 0° and 90° planes, respectively. In the E-plane (Fig. 5 a, b), the main lobe magnitudes are 14.6 dB(V/m) and 21.5 dB(V/m), with beam directions of 2.0° and 34.0° , and associated 3 dB beamwidths of 12.8° and 41.6° . The corresponding side-lobe levels are -2.1 dB and -9.0 dB. The H-plane patterns in Fig. 5 (c, d) exhibit main lobe magnitudes of -36.9 dB(A/m) and -30 dB(A/m), while maintaining the same beam directions, beamwidths, and side-lobe levels. The close agreement between the E-field and H-field patterns confirms stable radiation characteristics, good symmetry, and consistent polarization at the operating frequency.

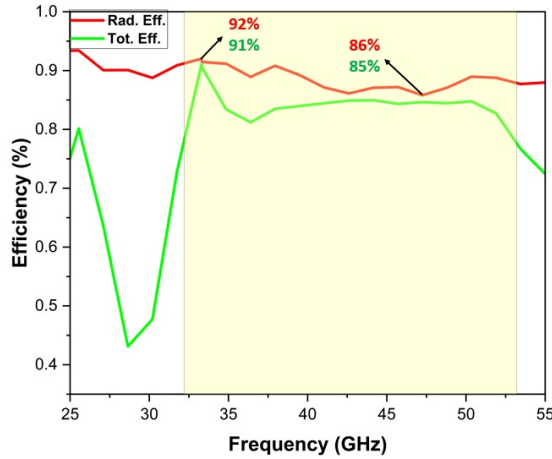


Fig. 3. Simulated efficiency graph of MIMO antenna.

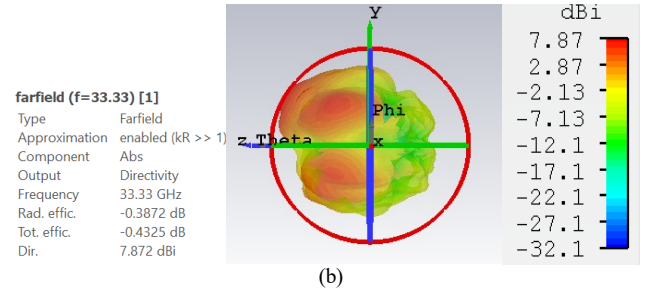
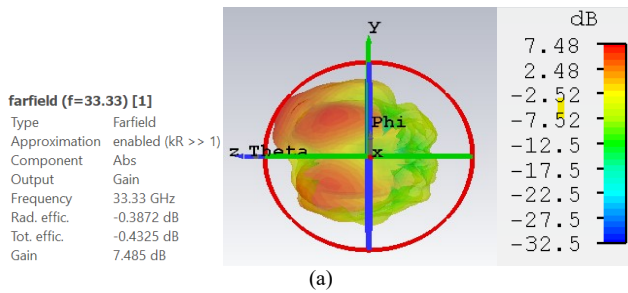


Fig. 4. 3D Radiation pattern with (a) gain and (b) directivity at 33.33 GHz.

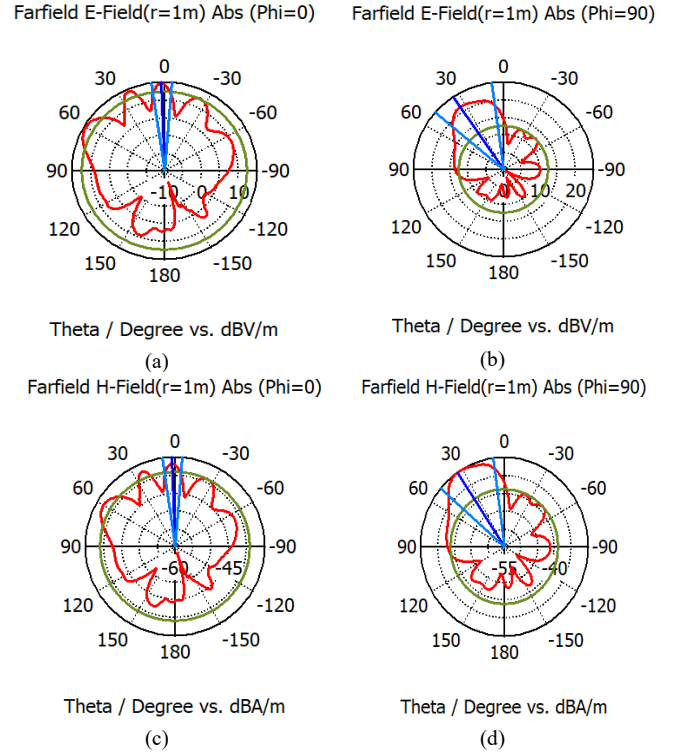


Fig. 5. (a) E-field at $\phi=0^\circ$, (b) E-field at $\phi=90^\circ$ (c) H-field at $\phi=0^\circ$ (d) H-field at $\phi=90^\circ$, all are plotted at frequency 33.33 GHz.

The MIMO performance of the proposed antenna is evaluated using the envelope correlation coefficient derived from far-field simulation results, as shown in Fig. 6. Across the entire operating band, the ECC remains extremely low, with a minimum value of 1.2875×10^{-6} . This indicates negligible correlation between the antenna elements, which are essential for effective diversity performance in 5G NR and D2D communication systems.

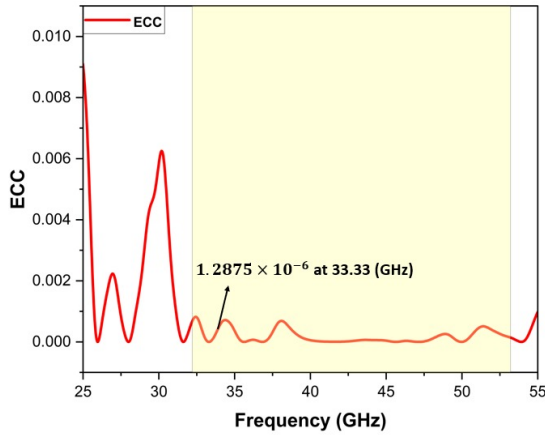


Fig. 6. ECC of MIMO antenna.

For practical measurements, ECC is commonly estimated from scattering parameters using the well-known relation [12]

$$\rho_e = \frac{|S_{11}^* S_{12} + S_{21}^* S_{22}|^2}{(1 - |S_{11}|^2 - |S_{21}|^2)(1 - |S_{22}|^2 - |S_{12}|^2)} \quad (2)$$

The simulated input impedance of the antenna is shown in Fig. 7. Ideally, the real part of the input impedance should be close to $50 \pm 2 \Omega$, while the imaginary part should approach $0 \pm 2 \Omega$. At 33.33 GHz, the antenna exhibits an input impedance of $Z_{in} = 51.88 + j1.84 \Omega$, which is in good agreement with the reflection coefficient results and confirms minimal signal reflection.

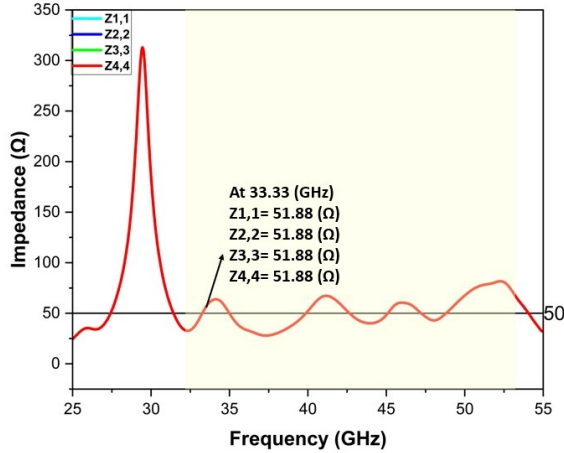


Fig. 7. Impedance of Z-matrix analysis in the real part.

The surface current distributions on the radiating patch and defected ground plane at 33.33 GHz are illustrated in Fig. 8. The maximum current density, reaching $126.901 \text{ A}^2/\text{m}$, is primarily concentrated along the edges of the radiating patch. The current propagates upward along the feed line and is guided toward the patch edges and diagonal slots. These slots modify the current paths, support the desired resonant mode, and direct energy toward the radiating edges. The defected ground plane further contributes to enhanced radiation by redistributing surface currents and suppressing mutual coupling. Overall, the observed current distribution confirms efficient energy guidance and strong resonance behavior of the proposed antenna.

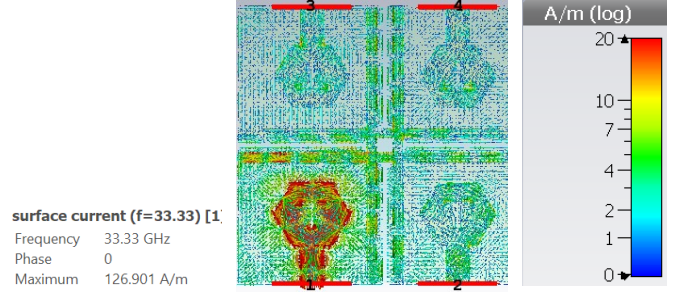


Fig. 8. Illustration of the surface current characteristics of the MIMO antenna at 33.33 GHz frequency for port 1.

IV. COMPARATIVE ANALYSIS

Table III compares the proposed antenna with several recently reported millimeter-wave MIMO antenna designs, with emphasis on key performance metrics relevant to 5G NR applications, including antenna size, operating bandwidth, isolation, gain, envelope correlation coefficient (ECC), and radiation efficiency.

It can be observed that some previously reported designs occupy relatively large physical areas, for example, up to $80 \times 80 \text{ mm}^2$, and offer wide frequency coverage in the range of 23–40 GHz. However, these designs typically exhibit moderate isolation levels (above -20 dB) and relatively low radiation efficiencies of around 70%. Other reported antennas achieve more compact dimensions and improved efficiency, but this is often accompanied by limitations in bandwidth or isolation performance [5], [10]. Broadband dual-band 28/38 GHz designs exhibit favorable gain and efficiency, but may require larger footprints due to geometry and integration constraints [8]. In contrast, some compact dual-band approaches offer a compromise between size and performance, though they may operate over a narrower frequency range [2].

By comparison, the proposed antenna achieves a favorable balance among compact size, wide bandwidth, and MIMO performance. With an overall footprint of $36 \times 36 \text{ mm}^2$, the antenna operates over 32.198–53.327 GHz, corresponding to an impedance bandwidth of 21.13 GHz. In addition, a high isolation level of -47.43 dB is obtained, along with a competitive peak gain of 8.71 dB and an ultra-low ECC of 1.2875×10^{-6} . The radiation efficiency remains high at approximately 92%. These combined characteristics indicate strong diversity performance and demonstrate that the proposed antenna is well-suited for compact 5G NR millimeter-wave and short-range D2D communication systems.

V. CONCLUSION

This paper has presented a compact four-element MIMO antenna designed for 5G New Radio (NR) millimeter-wave and device-to-device (D2D) communication applications. The proposed antenna occupies a small footprint of $36 \times 36 \text{ mm}^2$ and operates over 32.198–53.327 GHz, providing an impedance bandwidth of 21.13 GHz. The design achieves a high level of isolation of up to -47.43 dB while maintaining a competitive peak gain of 8.71 dB, and its performance has been validated through full-wave electromagnetic simulations. Prototype

fabrication and experimental measurements will be conducted in future work to further verify the simulated performance under practical operating conditions. In addition, the antenna exhibits an ultra-low envelope correlation coefficient of 1.2875×10^{-6} and a high radiation efficiency of approximately 92%, indicating strong MIMO diversity performance with minimal power loss. The combination of compact size, wideband operation, high isolation, and low correlation demonstrates a favorable trade-off compared with previously reported designs. These compact footprints and high isolation performance make the proposed antenna suitable for practical integration into compact 5G NR devices, including handheld terminals, portable hotspots, and short-range D2D communication platforms. The low-profile structure and wide operating bandwidth also support future miniaturized mmWave communication modules where space constraints and electromagnetic interference are critical design

concerns. In addition, the antenna exhibits an ultra-low envelope correlation coefficient of 1.2875×10^{-6} and a high radiation efficiency of approximately 92%, indicating strong MIMO diversity performance with minimal power loss. The combination of compact size, wideband operation, high isolation, and low correlation demonstrates a favorable trade-off compared with previously reported designs. These compact footprints and high isolation performance make the proposed antenna suitable for practical integration into compact 5G NR devices, including handheld terminals, portable hotspots, and short-range D2D communication platforms. The low-profile structure and wide operating bandwidth also support future miniaturized mmWave communication modules where space constraints and electromagnetic interference are critical design

TABLE III. COMPARISON OF THE PROPOSED MIMO ANTENNA WITH VARIOUS RELEVANT WORKS

Ref.	Total Size (L×W) mm ²	Operating frequency range (GHz)	BW (GHz)	Isolation (dB)	Max. Gain (dB)	ECC	Efficiency (%)
[2]	43.611×43.611	27.5–28.7, 36.8–38.8	1.2, 2.0	<−20, <−30	5.8 4.6	<0.0002, <0.0003	89, 92
[4]	80×80	23–40	17	>−20	12	<0.0014	70
[5]	47.4×32.5	36.83–40.0	3	−45	6.5	<0.001	>80
[10]	38.6×38.6	28, 38	2.52, 6.5	−15, −25	7.4/7.5	<0.0001	90.5
[8]	55×110	28/38	1.068, 1.431	−26	7.95, 8.27	2.46×10^{-5} , 7.65×10^{-5}	89.89, 88.25
[13]	70×16	37.7–40.3	2.6	<−25	7.5	<0.005	80
[14]	37.7×37.7	24.25–29.50	13.5	<−26	9.8	<0.002	88
[15]	28×28	36.5–48	11.5	<−18	9.8	<0.002	86
Proposed Design	36×36	32.198 – 53.327	21.13	−47.43	8.71	1.2875×10^{-6}	92

REFERENCES

- [1] A. A. Ibrahim, K. A. Mahmoud, and H. A. Elsayed, “Four-port 38 GHz MIMO antenna with high gain and isolation for 5G wireless networks,” *Sensors*, vol. 23, no. 7, Art. no. 3557, 2023.
- [2] B. Aghoutane, R. El Amrani, and H. Boussekine, “A novel dual-band high-gain four-port millimeter-wave MIMO antenna array for 28/37 GHz 5G applications,” *AEU-Int. J. Electron. Commun.*, vol. 145, Art. no. 154071, 2022.
- [3] E. G. Larsson, O. Edfors, F. Tufvesson, and T. L. Marzetta, “Massive MIMO for next generation wireless systems,” *IEEE Commun. Mag.*, vol. 52, no. 2, pp. 186–195, Feb. 2014.
- [4] D. A. Sehrai, A. Malik, and I. Qureshi, “A novel high-gain wideband MIMO antenna for 5G millimeter-wave applications,” *Electronics*, vol. 9, no. 6, Art. no. 1031, 2020.
- [5] D. A. Sehrai, A. Malik, and I. Qureshi, “Compact quad-element high-isolation wideband MIMO antenna for mm-wave applications,” *Electronics*, vol. 10, no. 11, Art. no. 1300, 2021.
- [6] H. M. Marzouk, M. I. Ahmed, and A. A. Shaalan, “A novel dual-band 28/38 GHz AFSL MIMO antenna for 5G smartphone applications,” in *J. Phys.: Conf. Ser.*, vol. 1447, no. 1, Art. no. 012025, 2020.
- [7] E. Al Abbas, H. A. Hassan, and A. Kaddour, “MIMO antenna system for multi-band millimeter-wave 5G and wideband 4G mobile communications,” *IEEE Access*, vol. 7, pp. 181916–181923, 2019.
- [8] H. M. Marzouk, M. I. Ahmed, and A. H. A. Shaalan, “Novel dual-band 28/38 GHz MIMO antennas for 5G mobile applications,” *Prog. Electromagn. Res. C*, vol. 93, pp. 103–117, 2019.
- [9] F. Alnemr, M. F. Ahmed, and A. A. Shaalan, “A compact 28/38 GHz circularly polarized MIMO antenna for 5G applications,” *J. Infrared Millim. Terahertz Waves*, vol. 42, no. 3, pp. 338–355, 2021.
- [10] W. T. Sethi, A. Malik, and I. Qureshi, “Pattern-diversity-based four-element dual-band MIMO patch antenna for 5G mmWave communication networks,” *J. Infrared Millim. Terahertz Waves*, vol. 45, no. 5, pp. 521–537, 2024.
- [11] 3GPP, “NR; Base station (BS) radio transmission and reception,” 3GPP TS 38.104, Release 17, 2021.
- [12] S. Blanch, J. Romeu, and I. Corbella, “Exact representation of antenna system diversity performance from input parameter description,” *Electron. Lett.*, vol. 39, no. 9, pp. 705–707, May 2003.
- [13] N. A. Abbasi *et al.*, “High-Isolation array antenna design for 5G Mm-Wave MIMO applications,” *Journal of Infrared Millimeter and Terahertz Waves*, vol. 46, no. 1, Dec. 2024, doi: 10.1007/s10762-024-01027-3.
- [14] H. Zakeri, G. Moradi, M. Alibakhshikenari, B. S. Virdee, C. H. See and S. Koziel, “Wideband High-Gain mm-wave MIMO Antenna Design for 5G IoT Applications,” in *IEEE Internet of Things Journal*, doi: 10.1109/JIOT.2026.3681922.
- [15] S. Khan *et al.*, “Highly Compact Wideband High-Gain Four-Element MIMO Antenna for 5G New Radio IoT,” in *IEEE Internet of Things Journal*, vol. 12, no. 12, pp. 22350–22365, 15 June15, 2025, doi: 10.1109/JIOT.2025.3551092.